Homework 1

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Some of the problems are collected from the examples in the book.

Problem 1

Let $\mathcal{C}_{n-1}(t)$ denote the set of all continuous polynomial functions defined on $[0, n-1]$ with constant slope in $[i-1, i]$, $i = 1, 2, \ldots, n-1$, n is an integer. With a vector $\alpha(t) \in \mathcal{C}_{n-1}(t)$, we associate an n-dimensional column vector $\alpha = [\alpha(0)\ \alpha(1)\ \ldots\ \alpha(n-1)]^T$.

(1) Show that the vectors $\alpha_i(t) \in \mathcal{C}_{n-1}(t)$, $i = 1, 2, \ldots, n$, are linearly independent if and only if so are the associated column vectors $\alpha_i \in \mathbb{R}^n$, $i = 1, 2, \ldots, n$.

(2) Show that $\alpha_i(t) \in \mathcal{C}_3(t)$, $i = 1, 2, 3, 4$ with

$$\alpha_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \quad \alpha_4 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

are linearly independent thus form a basis for $\mathcal{C}_3(t)$ and so do the vectors $\beta_i(t) \in \mathcal{C}_3(t)$, $i = 1, 2, 3, 4$ with

$$\beta_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \beta_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \beta_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \beta_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

(3) Find the transition matrix T that transforms the basis $[\alpha_1(t)\ \alpha_2(t)\ \alpha_3(t)\ \alpha_4(t)]$ to the basis $[\beta_1(t)\ \beta_2(t)\ \beta_3(t)\ \beta_4(t)]$.

(4) Find the representation of $\xi(t)$ with

$$\xi = [\xi(0)\ \xi(1)\ \xi(2)\ \xi(3)]^T$$

with respect to the basis $[\beta_1(t)\ \beta_2(t)\ \beta_3(t)\ \beta_4(t)]$.

Solution

(1) The nodal map $E(f) = [f(0), \ldots, f(n-1)]^T$ is linear. Any nodal values $v_0, \ldots, v_{n-1}$ uniquely determine each segment:

$$f(t) = (i-t)v_{i-1} + (t-i+1)v_i, \qquad t \in [i-1, i].$$

Thus E is a linear isomorphism, $\dim \mathcal{C}_{n-1}(t) = n$, and

$$\sum_{i=1}^{n} c_i \alpha_i(t) = 0 \quad \Longleftrightarrow \quad \sum_{i=1}^{n} c_i \alpha_i = 0.$$

This proves the equivalence of linear independence.

(2) Let $A_\alpha = [\alpha_1\ \alpha_2\ \alpha_3\ \alpha_4]$ and $A_\beta = [\beta_1\ \beta_2\ \beta_3\ \beta_4]$. Both are upper triangular with unit diagonal, so both determinants are one. By (1) and $\dim \mathcal{C}_3(t) = 4$, both lists of functions are bases.

(3) From $[\beta_1(t)\ \cdots\ \beta_4(t)] = [\alpha_1(t)\ \cdots\ \alpha_4(t)]T$, we obtain

$$\boxed{T = A_\alpha^{-1} A_\beta = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

(4) If y is the required coordinate column, then $\xi = A_\beta y$. Hence

$$\boxed{y = A_\beta^{-1}\xi = \begin{bmatrix} \xi(0) - \xi(1) \\ \xi(1) - \xi(2) \\ \xi(2) - \xi(3) \\ \xi(3) \end{bmatrix}}.$$

Problem 2

Please show that the vectors

$$A_1 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

are linearly independent and form a basis for $\mathbb{C}^{2\times 2}$, and find the presentation for the given matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

with respect to this basis.

Solution

Suppose $\sum_{i=1}^4 c_i A_i = 0$ and set $s = \sum_{i=1}^4 c_i$. Comparing entries gives $s - c_i = 0$ for $i = 1, 2, 3, 4$, so every $c_i = s$. Then $s = 4s$ forces $s = 0$, proving linear independence. Since $\dim_{\mathbb{C}} \mathbb{C}^{2\times 2} = 4$, these matrices form a basis.

$$A = \frac{-A_1 + 2A_2 + 2A_3 - A_4}{3}, \qquad \boxed{[A]_{(A_1,A_2,A_3,A_4)} = \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}^T}.$$

Problem 3

Let $\mathcal{C}_{n-1}(t)$ denote the set of all continuous polynomial functions defined on $[0, n-1]$ with constant slope in $[i-1, i]$, $i = 1, 2, \ldots, n-1$, n is an integer. Let $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s$ be a set of vectors of $\mathcal{V}$, the linear span of $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s$ is referred to as the set of all possible linear representations of these vectors

$$\text{span}\{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s\} = \{\boldsymbol{\alpha} : \boldsymbol{\alpha} = k_1\boldsymbol{\alpha}_1 + k_2\boldsymbol{\alpha}_2 + \cdots + k_s\boldsymbol{\alpha}_s,\ k_i \in \mathcal{F}\}.$$

In this problem, let $\alpha_1(t)$ and $\alpha_2(t) \in \mathcal{C}_3(t)$ with

$$\alpha_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathcal{U} = \text{span}\{\alpha_1(t), \alpha_2(t)\},$$

$\beta_1(t)$ and $\beta_2(t) \in \mathcal{C}_3(t)$ with

$$\beta_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \beta_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \mathcal{V} = \text{span}\{\beta_1(t), \beta_2(t)\}.$$

(1) The sum of $\mathcal{U}$ and $\mathcal{V}$ is defined as

$$\mathcal{U} + \mathcal{V} = \{\alpha = \alpha_1 + \alpha_2 : \alpha_1 \in \mathcal{U} \text{ and } \alpha_2 \in \mathcal{V}\}.$$

Please find out a basis for $\mathcal{U} + \mathcal{V}$ and its dimension;

(2) The intersect of $\mathcal{U}$ and $\mathcal{V}$ is defined as

$$\mathcal{U} \cap \mathcal{V} = \{\alpha : \alpha \in \mathcal{U} \text{ and } \alpha \in \mathcal{V}\}.$$

Please find out a basis for $\mathcal{U} \cap \mathcal{V}$ and its dimension.

Solution

By Problem 1, we may work with nodal vectors.

(1) We have $\beta_2 = \alpha_1 - \alpha_2 + \beta_1$. Moreover,

$$a\alpha_1 + b\alpha_2 + c\beta_1 = (a + b, b + c, c, a)^T = 0 \quad \Longrightarrow \quad a = b = c = 0.$$

Thus a basis and the dimension are

$$\boxed{\{\alpha_1(t), \alpha_2(t), \beta_1(t)\}}, \qquad \boxed{\dim(\mathcal{U} + \mathcal{V}) = 3}.$$

(2) Comparing components in $a\alpha_1 + b\alpha_2 = c\beta_1 + d\beta_2$ gives $a = d,\ b = c = -a$. The common vector is $a(\alpha_1 - \alpha_2)$, so

$$\boxed{\{\alpha_1(t) - \alpha_2(t)\}}, \qquad \boxed{\dim(\mathcal{U} \cap \mathcal{V}) = 1}.$$

Examples from the Textbook

Read the following examples before attempting Problem 4.

Example 1.23 (p. 21). Let $\mathcal{D} : \mathbb{R}_{n+1}[\lambda] \to \mathbb{R}_n[\lambda]$ be defined as

$$\mathcal{D}(f(\lambda)) = \frac{d}{d\lambda} f(\lambda) \qquad \forall f(\lambda) \in \mathbb{R}_{n+1}[\lambda].$$

Then from the linearity of derivative, it is readily to check that $\mathcal{D}$ is a linear transformation.

Example 1.24 (p. 22). Let $\mathcal{S} : \mathbb{R}_n[\lambda] \to \mathbb{R}_{n+1}[\lambda]$ be defined as

$$\mathcal{S}(f(\lambda)) = \int_0^\lambda f(\xi)\, d\xi \qquad \forall f(\lambda) \in \mathbb{R}_n[\lambda].$$

Then from the linearity of integration, it is readily to check that $\mathcal{S}$ is a linear transformation.

Example 1.26 (p. 25). Determine the matrix representation of the linear transformation $\mathcal{D} \in \mathcal{L}(\mathbb{R}_n[\lambda] \to \mathbb{R}_{n-1}[\lambda])$,

$$\mathcal{D}(f(\lambda)) = \frac{d}{d\lambda} f(\lambda)$$

with respect to the bases $[1\ \lambda\ \cdots\ \lambda^n]$ and $[1\ \lambda\ \cdots\ \lambda^{n-1}]$.

Example 1.27 (p. 26). Define the linear transformation $\mathcal{S} \in \mathcal{L}(\mathbb{R}_n[\lambda] \to \mathbb{R}_{n+1}[\lambda])$ as

$$\mathcal{S}(f(\lambda)) = \int_0^\lambda f(\xi)\, d\xi \qquad \forall f(\lambda) \in \mathbb{R}_n[\lambda].$$

Determine the matrix representation with respect to the basis $[1\ \lambda\ \cdots\ \lambda^n]$ and $[1\ \lambda\ \cdots\ \lambda^{n+1}]$.

Problem 4

Denote by $\mathbb{R}_{n-1}(\lambda)$ the set of all polynomials with real coefficients and degrees not exceeding $n-1$. With respect to the following addition and scalar product operations,

$$f_1(\lambda) = \sum_{i=0}^{n-1} f_i^{(1)} \lambda^i, \qquad f_2(\lambda) = \sum_{i=0}^{n-1} f_i^{(2)} \lambda^i$$

$$(f_1 + f_2)(\lambda) = \sum_{i=0}^{n-1} \big(f_i^{(1)} + f_i^{(2)}\big) \lambda^i$$

$$(k f_1)(\lambda) = k \sum_{i=0}^{n-1} f_i^{(1)} \lambda^i$$

(1) Let a linear transformation $\mathcal{L}_\alpha : \mathbb{R}_{n-1}(\lambda) \to \mathbb{R}_{n-1}(\lambda)$ be defined as

$$\mathcal{L}_\alpha[f(\lambda + \alpha)] = f(\lambda), \qquad f(\lambda + \alpha),\ f(\lambda) \in \mathbb{R}_{n-1}(\lambda),$$

where α is real constant number, show that $\mathcal{L}_\alpha$ is a linear transformation;

(2) Find out the matrix representation L for the linear transformation $\mathcal{L}_\alpha$ with respect to the basis $1, \lambda + \alpha, (\lambda + \alpha)^2, \ldots, (\lambda + \alpha)^{n-1}$, i.e., the matrix L satisfying

$$\begin{aligned} &\mathcal{L}_\alpha\big([1, \lambda + \alpha, (\lambda + \alpha)^2, \ldots, (\lambda + \alpha)^{n-1}]\big) \\ &\quad \triangleq [1, \lambda + \alpha, (\lambda + \alpha)^2, \ldots, (\lambda + \alpha)^{n-1}]L. \end{aligned}$$

Solution

(1) Set $g(\lambda) = f(\lambda + \alpha)$. Then $(\mathcal{L}_\alpha g)(\lambda) = g(\lambda - \alpha)$, and shifting does not increase degree. For any $a, b \in \mathbb{R}$,

$$\begin{aligned} \big(\mathcal{L}_\alpha(a g_1 + b g_2)\big)(\lambda) &= a g_1(\lambda - \alpha) + b g_2(\lambda - \alpha) \\ &= a(\mathcal{L}_\alpha g_1)(\lambda) + b(\mathcal{L}_\alpha g_2)(\lambda). \end{aligned}$$

Thus $\mathcal{L}_\alpha$ is linear.

(2) Let $b_j(\lambda) = (\lambda + \alpha)^j$, $0 \le j \le n-1$. Then

$$\mathcal{L}_\alpha b_j = \lambda^j = \sum_{i=0}^{j} \binom{j}{i} (-\alpha)^{j-i} b_i.$$

Placing these coordinates in columns gives

$$\boxed{L_{i+1,j+1} = \begin{cases} \binom{j}{i}(-\alpha)^{j-i}, & 0 \le i \le j \le n-1, \\ 0, & 0 \le j < i \le n-1. \end{cases}}$$